

POC 07: Digital Media Platform - Kubernetes, S3, Jenkins And Monitoring

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Role: **AWS DevOps Engineer**

Public-safe name: **Digital Media Platform**

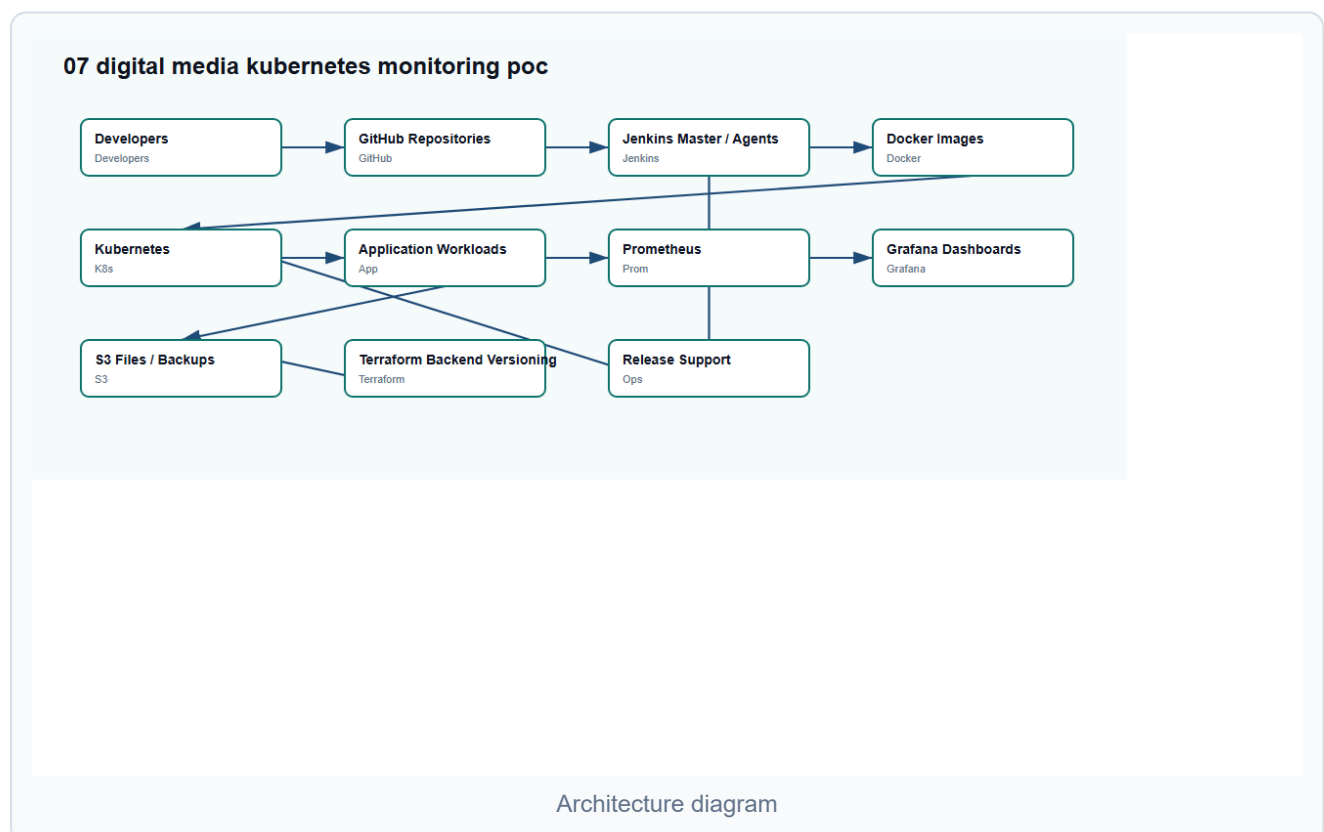
Confidentiality Note

This POC excludes private repository URLs, environment values, credentials, IP addresses, dashboard links, and client-specific deployment details.

POC Objective

Create a DevOps support blueprint for application availability, monitoring, S3 backup storage, Terraform backend versioning, Jenkins master-agent performance, GitHub repository maintenance, and Kubernetes-based container orchestration.

Support And Delivery Diagram



What I Worked On

- â€¢ Supported application availability and performance improvement activities.
- â€¢ Monitored build machines, deployment machines, and application environments using Prometheus and Grafana.
- â€¢ Created S3 buckets for files and backups.
- â€¢ Used Terraform backend versioning concepts.
- â€¢ Supported continuous delivery for releases.
- â€¢ Implemented Jenkins master-agent structure to improve Jenkins performance.
- â€¢ Maintained GitHub repositories and supported Kubernetes orchestration for Docker containers.

POC Implementation Scope

- â€¢ Jenkins master-agent pipeline pattern.
- â€¢ Kubernetes deployment and service manifests.
- â€¢ Prometheus/Grafana monitoring notes.
- â€¢ S3 backup bucket structure and lifecycle considerations.
- â€¢ Terraform backend versioning notes.
- â€¢ GitHub repository maintenance checklist.

Suggested Repository Structure

```
digital-media-k8s-monitoring-poc/  
|-- README.md  
|-- k8s/  
|   |-- deployment.yaml  
|   `-- service.yaml  
|-- jenkins/  
|   `-- Jenkinsfile  
|-- monitoring/  
|   |-- prometheus-notes.md  
|   `-- grafana-dashboard-notes.md  
|-- terraform-backend/  
|   `-- backend-example.hcl  
`-- docs/  
    |-- repository-maintenance.md  
    `-- release-support.md
```

Validation Steps

- Validate Kubernetes manifests.

- Review Jenkins master-agent pipeline stages.
- Confirm S3 backup/lifecycle plan.
- Review monitoring dashboard checklist.
- Confirm repository maintenance and release support runbooks.

Expected Outcome

- â€¢ Better release and support readiness.
- â€¢ Improved Jenkins performance pattern.
- â€¢ Clear Kubernetes deployment and monitoring documentation.
- â€¢ Structured S3 backup and Terraform backend planning.

Technologies

AWS S3, Terraform, Jenkins, GitHub, Kubernetes, Docker, Prometheus, Grafana, CI/CD, release support.